Robotics Basics - Sensors, Actuators, and Frames

Course: Introduction to Robotics (ROB150)
Instructor: Prof. Michael Brown
Added: 2025-10-03 08:45:00

Summary

Basic robotics note covering robot components, sensors, actuators, coordinate frames, and feedback loops.

Robot components

- A robot combines mechanical structure, sensors, actuators, control software, and a power system.
- The controller reads sensor data, decides the next action, and sends commands to actuators.

Sensors and actuators

- Sensors measure the environment or the robot state.
- Actuators create motion, such as wheel rotation, arm movement, or gripper force.

Coordinate frames

- A coordinate frame defines where positions and directions are measured from.
- A mobile robot may use a world frame, robot frame, and sensor frame at the same time.

Quick reference

Component	Example	Purpose
Sensor	Ultrasonic range sensor	Measures distance
Actuator	DC motor	Creates wheel motion
Controller	Microcontroller	Runs control logic
Frame	Robot coordinate frame	Defines local direction